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# Odd-Cycle-Free Latin Squares

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# Odd-Cycle-Free Latin Squares: Products, Congruence Obstructions, and an Order-8 Census

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## Abstract

For each pair of rows, columns, or symbols of a Latin square, the corresponding two-line permutation is fixed-point-free. We call a view failing when one of these permutations has a nontrivial odd cycle, and call the square FFF when none of its three views fails. Such cycles support Latin trades. We prove the exact three-parity sign table: odd row- and column-view trades reverse the conventional row-column sign, while odd symbol-view trades preserve it and reverse the two parastrophic companion signs.

We prove that odd-order squares are TTT, characterize FFF group tables, and give a basis-family closure criterion for group isotopy. Failure patterns are monotone from Latin subsquares and are preserved exactly through congruences with two-element blocks; odd block size forces TTT. Consequently every FFF quasigroup of order  $2p$ , for odd prime  $p$ , is congruence-simple in every isotope. A twisted binary extension gives an explicit non-group-isotopic FFF loop of order 8. Nonuniversal binary line dependencies are in bijection with half-order subsquares and occur in complementary four-packs. Hence an FFF square of order 2 modulo 4, greater than 2, has maximal binary line-incidence rank. Specializing known 3-net character theory, nonuniversal prime-field line dependencies are balanced cyclic quotient homotopies. Hence at order  $2p$ , for odd prime  $p$ , one F view forces a saturated integer line-incidence lattice.

For direct products we prove

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K)$$

componentwise, and show that  $L \times K$  is group-isotopic exactly when both factors are. An exhaustive reproducible scan of the 283657 order-8 main classes finds 230 FFF classes, of which 225 are not group-isotopic; all eight patterns occur. Exact generation also shows that no order-6 FFF square exists. Hence every order  $2^m$ ,  $m \geq 3$ , realizes all eight patterns and contains a non-group-isotopic FFF square. These results motivate, but do not prove, the conjecture that nontrivial FFF squares exist exactly in power-of-two orders.

**Keywords.** Latin square; quasigroup congruence; Latin subsquare; Latin trade; direct product; computational enumeration.

**2020 Mathematics Subject Classification.** Primary 05B15; Secondary 20N05, 05C70.

## Evidence convention

Unlabelled lemmas, propositions, and theorems below are proved mathematically from the stated definitions and cited results. Statements labelled *Computational Theorem* are exhaustive finite computations with frozen inputs, executable code, commands, hashes, and independent validation reports. Any corollary that explicitly invokes a Computational Theorem inherits that computational dependency. These results are not presented as formal proof-assistant certificates. The complete artifact map is given in Section 7.

The trade-parity convention is explicit: the conventional row-column sign and its two parastrophic companions are distinguished. No symbol-view trade is claimed to reverse the conventional sign merely by parastrophy.

# 1 Introduction

An odd cycle in the permutation induced by two lines of a Latin square supports a simple trade: swap the two lines on that cycle. The move preserves the Latin property. In the row and column views it reverses the conventional row–column Latin sign; in the symbol view it preserves that sign and reverses the two parastrophically transported signs. This local operation is therefore a natural primitive in parity-based approaches related to the Alon–Tarsi setting, but its three sign conventions must be kept distinct. Bérczi’s experimental program studies such local corrections as ingredients for global constructions [Bér26]; it does not supply the theorems proved here.

We isolate the underlying existence question. For each of the row, column, and symbol views, ask whether some pair of lines induces a permutation with a nontrivial odd cycle. The three answers form a pattern in  $\{F, T\}^3$ . A square is FFF when all three answers are negative. Equivalently, every union of two link factors in the tripartite representation has component lengths divisible by four.

Our first contribution is structural. Every odd-order Latin square is TTT, whereas the Cayley table of a finite group is FFF exactly when the group is a 2-group. A basis-family closure criterion detects group isotopy. We then show that failure patterns are monotone from Latin subsquares and analyze quasigroup congruences: odd block size forces TTT, while two-element blocks preserve the quotient pattern exactly. Thus a hypothetical FFF quasigroup of order  $2p$ , for odd prime  $p$ , must be congruence-simple in every isotope. This excludes all congruence-imprimitive constructions at order 10 but does not by itself decide the simple case.

The remaining binary line-incidence defect also has an exact subsquare meaning. Nonuniversal vectors in the left binary line kernel are in bijection with half-order subsquares, and the quotient classes group them into complementary four-packs. This yields a closed counting formula and forces maximal binary line rank for every FFF order congruent to 2 modulo 4 and greater than 2. At order 10 this is necessary, not sufficient.

More generally, Moorhouse’s 3-net character formula specializes as follows: a nonuniversal line dependency over a prime field is a balanced cyclic quotient homotopy, and after isotopy its fibres form a quasigroup congruence. Consequently every Smith torsion prime divides the order. For order  $2p$ , with  $p$  odd prime, the congruence obstruction shows that even one F view forces a saturated integer line-incidence lattice.

The group construction is not the whole phenomenon. A twisted binary extension of  $C_4$  gives an explicit non-group-isotopic FFF loop of order 8. Moreover, direct products satisfy the exact componentwise formula

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K),$$

and a product is group-isotopic exactly when both factors are. These theorems turn one order-8 example into infinite power-of-two families.

Our computational contribution is exhaustive. McKay, Meynert, and Myrvold enumerated the main classes of small Latin squares [MMM07]; the associated ANU file contains one representative of each of the 283657 order-8 main classes [McK]. Our scan finds exactly 230 FFF main classes. Only 5 are group-isotopic and 225 are not, and all eight ordered failure patterns occur. Independently, fresh exact-cover generation of all 9408 reduced order-6 squares finds no FFF square and matches the official ANU set exactly.

Combining theory and computation proves that every order  $2^m$ ,  $m \geq 3$ , realizes all eight patterns and contains a non-group-isotopic FFF square. We do not claim that the product construction is injective on isotopy or main classes. We also do not prove the

Alon–Tarsi conjecture. The accompanying complete research report proves by exhaustive master searches that order 10 has no FFF square; that computation is outside this compact paper’s theorem scope. The resulting open conjecture is narrower: nontrivial FFF Latin squares may exist exactly in power-of-two orders.

The longer companion research report in the repository develops additional cycle-parity, subset-action, primitive-group, and flag-surface machinery and proves nonexistence at order 10. None of that machinery is needed for the results in this compact paper.

## 2 Definitions and invariance

Let  $X$  be a set of size  $n$ . A *Latin square* on  $X$  is a map  $L: X \times X \rightarrow X$  such that every row and every column is a bijection. For  $a, c, u \in X$ , define the row, column, and symbol-line bijections

$$r_a(c) = L(a, c), \quad q_c(a) = L(a, c), \quad s_u(c) = \text{the unique } a \text{ such that } L(a, c) = u.$$

For two lines in the corresponding view, put

$$\rho_{a,b} = r_b^{-1} r_a, \quad \kappa_{c,d} = q_d^{-1} q_c, \quad \sigma_{u,v} = s_v^{-1} s_u.$$

All three permutations act on an  $n$ -element complementary coordinate set.

**Lemma 2.1.** *If the two lines are distinct, the corresponding induced permutation is fixed-point-free.*

*Proof.* A fixed point of  $\rho_{a,b}$  would give  $L(a, c) = L(b, c)$  in one column, contrary to the Latin property. The column case is identical. A fixed point of  $\sigma_{u,v}$  would say that the same cell contains the two distinct symbols  $u$  and  $v$ .  $\square$

**Definition 2.2.** A view of  $L$  *fails* if one of its induced two-line permutations has an odd cycle of length greater than one. Write

$$\text{pat}(L) = (\text{row}(L), \text{col}(L), \text{sym}(L)) \in \{F, T\}^3,$$

where  $T$  means that the view fails. A square is *FFF* if  $\text{pat}(L) = \text{FFF}$ .

The terminology “fails” is chosen from the perspective of the local-trade program: a  $T$  view supplies an odd-cycle trade, while an  $F$  view does not.

**Definition 2.3.** Fix two rows  $a \neq b$  and a union  $C$  of cycles of  $\rho_{a,b}$ . The associated *two-line trade* swaps the entries in rows  $a, b$  in precisely the columns in  $C$ . Column- and symbol-view trades are defined by permuting the coordinate roles.

**Proposition 2.4.** *A two-line trade produces another Latin square. Define the three line-sign products*

$$R(L) = \prod_a \text{sgn}(r_a), \quad C(L) = \prod_c \text{sgn}(q_c), \quad S(L) = \prod_u \text{sgn}(s_u),$$

and put

$$\epsilon_{RC} = RC, \quad \epsilon_{RS} = RS, \quad \epsilon_{CS} = CS.$$

The first is the conventional row-column Latin sign, while the other two are its parastrophic companions. If a trade has support size  $k$  and  $\delta = (-1)^k$ , then its sign ratios are

| trade view | $R'/R$   | $C'/C$   | $S'/S$   | $\epsilon'_{RC}/\epsilon_{RC}$ | $\epsilon'_{RS}/\epsilon_{RS}$ | $\epsilon'_{CS}/\epsilon_{CS}$ |
|------------|----------|----------|----------|--------------------------------|--------------------------------|--------------------------------|
| row        | 1        | $\delta$ | $\delta$ | $\delta$                       | $\delta$                       | 1                              |
| col        | $\delta$ | 1        | $\delta$ | $\delta$                       | 1                              | $\delta$                       |
| sym        | $\delta$ | $\delta$ | 1        | 1                              | $\delta$                       | $\delta$                       |

Consequently an odd row- or column-view trade reverses the conventional  $\epsilon_{RC}$ , whereas an odd symbol-view trade preserves  $\epsilon_{RC}$  and reverses  $\epsilon_{RS}$  and  $\epsilon_{CS}$ .

*Proof.* Write  $\pi = \rho_{a,b}$ . Since  $r_a(c) = r_b(\pi(c))$  and  $C$  is  $\pi$ -invariant, the two rows contain the same symbol set on  $C$ . Swapping those entries therefore preserves both row symbol sets, while each affected column merely swaps two entries. Thus the result is Latin.

Extend  $\pi|_C$  by the identity outside  $C$ . The two changed row permutations are obtained by precomposition with  $\pi$  and  $\pi^{-1}$ , so their sign changes cancel and  $R'/R = 1$ . Each of the  $|C|$  affected column permutations is composed with a transposition, so  $C'/C = (-1)^{|C|}$ . The same support contains the same  $|C|$  symbols in the two rows; each associated symbol-position permutation swaps those two rows, giving  $S'/S = (-1)^{|C|}$ . This proves the first table row. Exchanging rows and columns proves the second.

For a symbol-view trade, each support column swaps the two selected symbols, and the cycle relation pairs the affected cells in exactly the same number of rows. Hence  $R'/R = C'/C = (-1)^{|C|}$ . The two changed symbol-position permutations differ from their predecessors by mutually inverse permutations on the support, so their sign changes cancel and  $S'/S = 1$ . Multiplying the appropriate line-sign ratios proves the last three columns.  $\square$

**Proposition 2.5** (Tripartite link formulation). *Regard the cells of  $L$  as the triangle decomposition*

$$\{\{a, c, L(a, c)\} : a, c \in X\}$$

*of the complete tripartite graph on the row, column, and symbol copies of  $X$ . The links of the vertices in each part form a 1-factorization between the other two parts. An induced two-line cycle of length  $\ell$  corresponds bijectively to an alternating cycle of length  $2\ell$  in the union of the two associated link factors. Hence  $L$  is FFF if and only if every two-factor union in all three link factorizations has only cycle lengths divisible by four.*

*Furthermore, the total number of 4-cycles among the two-factor unions is the same in all three views and equals the number of intercalates of  $L$ .*

*Proof.* For a row  $a$ , its incident triangles give the perfect matching  $M_a = \{\{c, r_a(c)\} : c \in X\}$  between columns and symbols. For distinct rows  $a, b$ , alternately following an edge of  $M_a$  and an edge of  $M_b$  sends a column  $c$  to  $r_b^{-1}r_a(c) = \rho_{a,b}(c)$ . Thus an  $\ell$ -cycle of  $\rho_{a,b}$  is exactly one alternating  $2\ell$ -cycle of  $M_a \cup M_b$ . Cyclically permuting the three coordinate classes proves the column and symbol statements and the FFF equivalence.

A 4-cycle in  $M_a \cup M_b$  is exactly a 2-cycle of  $\rho_{a,b}$ , hence a 2-by-2 Latin subsquare on two rows, two columns, and two symbols. This is an intercalate. The same intercalate is counted once by its row pair, once by its column pair, and once by its symbol pair, proving the three equalities.  $\square$

**Lemma 2.6** (Global three-view sign identity). *For every Latin square of order  $n$ ,*

$$R(L)C(L)S(L) = (-1)^{\binom{n}{2}}.$$

*Proof.* Give each coordinate copy of  $X$  the same fixed order, and let  $\tau(x, y) = (y, x)$  on  $X^2$ . The  $\binom{n}{2}$  off-diagonal pairs are the transpositions of  $\tau$ , so  $\text{sgn}(\tau) = (-1)^{\binom{n}{2}}$ .

The row-entry map  $A(r, c) = (r, L(r, c))$  is block diagonal by rows and has sign  $R(L)$ . The map  $B(r, c) = (c, L(r, c))$  is a coordinate transposition followed by the block-diagonal family of column permutations, so its sign is  $(-1)^{\binom{n}{2}}C(L)$ . Likewise,  $D(c, s) = (s, s_s(c))$  has sign  $(-1)^{\binom{n}{2}}S(L)$ . Since  $A = \tau \circ D \circ B$ , taking signs gives

$$R(L) = (-1)^{3\binom{n}{2}}S(L)C(L) = (-1)^{\binom{n}{2}}S(L)C(L),$$

which is equivalent to the claimed identity.  $\square$

For later comparison, let  $C_{v,k}$  denote the total number of  $k$ -cycles among all induced line-pair permutations in view  $v$ .

**Proposition 2.7** (Aggregate cycle-profile baseline). *Let  $n$  be even and let  $p$  be prime. If*

$$\sum_{k=2}^n a_k x_k + b = 0 \pmod{p}$$

*for every cycle-count vector  $(x_2, \dots, x_n)$  of a fixed-point-free permutation of degree  $n$ , then, separately for each view  $v$ ,*

$$\sum_{k=2}^n a_k C_{v,k} + \binom{n}{2} b = 0 \pmod{p}.$$

*Moreover,*

$$C_{\text{row},2} = C_{\text{col},2} = C_{\text{sym},2}$$

*and*

$$\sum_{v \in \{\text{row}, \text{col}, \text{sym}\}} \sum_{\substack{2 \leq k \leq n \\ k \text{ even}}} C_{v,k} = \frac{n(n-1)}{2} \pmod{2}.$$

*Proof.* The aggregate profile in one view is the sum of the  $\binom{n}{2}$  fixed-point-free line-pair profiles, proving the first identity. The second identity is the intercalate bijection from Proposition 2.5. For the last identity, the sign of a fixed-point-free permutation of even degree is  $(-1)^e$ , where  $e$  is its number of even cycles. Multiplying relative-permutation signs over all line pairs in a view gives the product of its line signs. Lemma 2.6 gives the product over the three views as  $(-1)^{n(n-1)/2}$ , which proves the congruence.  $\square$

**Proposition 2.8** (Line-pair sign cut). *Fix one view and write its line bijections as  $\lambda_i$ . For  $P_{ij} = \lambda_j^{-1} \lambda_i$ , the pairs for which  $P_{ij}$  is odd form a cut of the complete graph on the lines. If  $n$  is even and the view is  $F$ , then the pairs for which  $P_{ij}$  has an odd number of cycles form the same cut. Consequently, for even  $n > 2$ , no view can have every induced line-pair permutation equal to one  $n$ -cycle.*

*Proof.* Put  $\varepsilon_i = \text{sgn}(\lambda_i)$ . Since sign is a homomorphism,

$$\text{sgn}(P_{ij}) = \varepsilon_i \varepsilon_j.$$

Thus  $P_{ij}$  is odd exactly when the endpoint signs differ. These pairs are the edges of the cut between positive and negative lines.

In an  $F$  view of even degree, every cycle of  $P_{ij}$  has even length and therefore negative sign. Hence  $\text{sgn}(P_{ij}) = (-1)^{c(P_{ij})}$ , where  $c(P_{ij})$  is the number of cycles. Finally, if all relative permutations were  $n$ -cycles, every edge would be negative. A triangle of three lines would then cross the sign cut three times, whereas every cut meets a cycle evenly.  $\square$

An *isotopy* independently permutes rows, columns, and symbols. A *parastrophy* permutes the three coordinate roles. Their combined orbits are the *main classes* (also called species).

**Proposition 2.9.** *Under isotopy, each of the three view bits is invariant. Under parastrophy, the three bits are permuted. Consequently, being FFF and having at least one failing view are main-class invariants.*

*Proof.* Let an isotopy act by permutations  $\alpha, \beta, \gamma$  of rows, columns, and symbols. The transformed row maps satisfy  $r'_{\alpha(a)} = \gamma r_a \beta^{-1}$ , and hence

$$(r'_{\alpha(b)})^{-1} r'_{\alpha(a)} = \beta(r_b^{-1} r_a) \beta^{-1}.$$

Thus row cycle lengths are unchanged. The column calculation conjugates by  $\alpha$ , and the symbol calculation conjugates by  $\beta$ . Parastrophy only permutes which coordinate pair is called row, column, or symbol.  $\square$

### 3 Elementary structural results

#### 3.1 Odd orders and group tables

**Proposition 3.1.** *Every Latin square of odd order  $n > 1$  has pattern TTT.*

*Proof.* By Lemma 2.1, every induced permutation between two distinct lines is fixed-point-free on an odd set. If all of its nontrivial cycles were even, their lengths would have even sum. Hence it has an odd cycle of length greater than one. This argument applies separately to any pair of distinct rows, columns, and symbols.  $\square$

Let  $G$  be a finite group and let  $L_G(x, y) = xy$  be its Cayley table.

**Proposition 3.2.** *The row permutation induced by rows  $a, b$  of  $L_G$  is left multiplication by  $b^{-1}a$ . The column permutation induced by columns  $c, d$  is right multiplication by  $cd^{-1}$ . The symbol permutation induced by symbols  $u, v$  is right multiplication by  $u^{-1}v$ . Each cycle length is therefore the order of the corresponding group element.*

*Proof.* The row identity is  $r_b^{-1} r_a(x) = b^{-1} a x$ . Likewise  $q_d^{-1} q_c(x) = x c d^{-1}$ . Finally,  $s_u(c) = u c^{-1}$  and  $s_v^{-1}(a) = a^{-1} v$ , whence  $s_v^{-1} s_u(c) = c u^{-1} v$ . Left or right multiplication by  $g$  has orbits equal to cosets of  $\langle g \rangle$ , all of length  $\text{ord}(g)$ .  $\square$

**Corollary 3.3.** *The Cayley table of a finite group  $G$  is FFF if and only if  $G$  is a 2-group.*

*Proof.* If  $G$  is a 2-group, every nonidentity element has even order, so Proposition 3.2 gives no nontrivial odd cycle. Conversely, if  $|G|$  has an odd prime divisor, Cauchy's theorem supplies a nonidentity element of odd order; the corresponding induced permutations give a failing view.  $\square$

#### 3.2 A closure criterion for group isotopy

Fix a base row  $b$  of a Latin square  $L$  and define

$$H_L(b) = \{r_b^{-1} r_x : x \in X\} \subseteq \text{Sym}(X).$$

**Proposition 3.4** (Basis-family closure criterion). *A Latin square  $L$  is isotopic to a group table if and only if  $H_L(b)$  is closed under composition. This condition is independent of the chosen base row.*

*Proof.* Suppose first that  $H = H_L(b)$  is closed. It contains the identity, has  $|X|$  distinct elements, and is a finite submonoid of a symmetric group; hence it is a group. For each  $z \in X$ , the values  $(r_b^{-1}r_x)(z)$ , as  $x$  varies, are all distinct, since a column of  $L$  contains every symbol once. Thus  $H$  acts regularly on  $X$ .

Fix  $z_0 \in X$ . Relabel rows by  $x \mapsto h_x = r_b^{-1}r_x$ . Use the regular action to identify every column  $z$  with the unique  $h_z \in H$  satisfying  $h_z(z_0) = z$ , and identify every relabeled symbol  $r_b^{-1}(u)$  in the same way. The entry in row  $h_x$  and column  $h_z$  is represented by the unique element sending  $z_0$  to  $h_x h_z(z_0)$ , namely  $h_x h_z$ . The resulting isotope is therefore the Cayley table of  $H$ .

Conversely, for a group table the family at base row  $b$  is the full left regular representation  $\{\lambda_{b^{-1}x} : x \in G\}$  and is closed. Isotopy conjugates the family and relabels its members, preserving closure. The same argument for any base row proves base independence.  $\square$

The same criterion can equivalently be written in the column or symbol view, by parastrophy.

## 4 Subsquare and congruence obstructions

The FFF property is hereditary for Latin subsquares in a stronger, componentwise sense.

**Theorem 4.1** (Subsquare monotonicity). *If  $M$  is a Latin subsquare of  $L$ , then*

$$\text{pat}(M) \leq \text{pat}(L)$$

*componentwise, with  $F < T$ .*

*Proof.* Let the rows, columns, and symbols of  $M$  be  $R_0, C_0, S_0$ . For  $a, b \in R_0$ , the ambient row permutation  $r_b^{-1}r_a$  preserves  $C_0$ : if  $c \in C_0$ , then  $L(a, c) \in S_0$ , and row  $b$  contains this symbol at a unique column of  $C_0$ . Its restriction is precisely the induced row permutation of  $M$ . Thus every odd row cycle of  $M$  is also an odd row cycle of  $L$ . Interchanging rows and columns proves the column statement.

For  $u, v \in S_0$  and  $c \in C_0$ , the row  $s_u(c)$  lies in  $R_0$ . In that row, the unique occurrence of  $v$  lies in a column  $d \in C_0$ . Hence  $s_v^{-1}s_u$  preserves  $C_0$  and restricts to the symbol permutation of  $M$ . This proves the symbol statement.  $\square$

**Corollary 4.2.** *Every Latin subsquare of an FFF square is FFF. In particular, an FFF square contains no subsquare of odd order greater than one.*

*Proof.* The first assertion is Theorem 4.1 with  $\text{pat}(L) = \text{FFF}$ . The second follows from Proposition 3.1.  $\square$

Together with Computational Theorem 6.1, the same implication excludes order-6 subsquares; that additional exclusion inherits the computational status of the order-6 theorem.

There is an exact linear-algebraic description of all half-order subsquares. Let  $M_L$  be the  $3n$ -by- $n^2$  binary incidence matrix whose rows are the row, column, and symbol lines and whose columns are the occupied cells of  $L$ . Let

$$U = \{(\alpha \mathbf{1}_R, \beta \mathbf{1}_C, \gamma \mathbf{1}_S) : \alpha + \beta + \gamma = 0\} \subseteq \ker M_L^\top$$

and put

$$\delta_2(L) = (3n - 2) - \text{rank}_{\mathbb{F}_2} M_L.$$

**Theorem 4.3** (Binary line kernel and half-order subsquares). *Individual embedded half-order Latin subsquares of  $L$  are in bijection with*

$$\ker M_L^T \setminus U.$$

*Equivalently, the nonzero classes in  $\ker M_L^T/U$  parametrize their complementary four-packs. Thus their number is exactly*

$$4(2^{\delta_2(L)} - 1). \quad (1)$$

*If  $n = 2m$ , where  $m > 1$  is odd, then any one F-view forces*

$$\text{rank}_{\mathbb{F}_2} M_L = 3n - 2. \quad (2)$$

*In particular, (2) holds for every FFF square of order congruent to 2 modulo 4 and greater than 2.*

*Proof.* A left-kernel vector is a triple  $(a, b, c)$  satisfying

$$a_r + b_j + c_{L(r,j)} = 0. \quad (3)$$

If one coordinate, say  $a$ , is constant, then fixing a column and varying its rows shows that  $c$  is constant; fixing a row then makes  $b$  constant. Hence every vector outside  $U$  has all three coordinates nonconstant.

Fix a row  $r$ . Since that row permutes the symbols, (3) identifies the multiset of  $c$ -values with that of  $a_r + b$ . Choosing rows with  $a_r = 0$  and  $a_r = 1$  gives

$$\text{wt}(c) = \text{wt}(b) = n - \text{wt}(b).$$

Thus  $n = 2m$  and  $b, c$  each have weight  $m$ . The analogous fixed-column argument gives  $\text{wt}(a) = m$ . For their binary fibres  $A_x, B_y, C_z$ , equation (3) now gives

$$L(A_x, B_y) \subseteq C_{x+y} \quad (x, y \in \mathbb{F}_2).$$

Every set has size  $m$ ; Latinness makes each of the four displayed blocks an order- $m$  Latin subsquare.

Adding an element of  $U$  complements zero or two of the three partitions and selects a different block in the same four-pack. Conversely, if  $R_0 \times C_0$  is a half-order subsquare with symbol set  $S_0$ , row and column counting forces the complementary pattern

$$R_0 \times C_0, R_1 \times C_1 \longrightarrow S_0, \quad R_0 \times C_1, R_1 \times C_0 \longrightarrow S_1.$$

The three complement indicators satisfy (3), and the chosen block recovers them uniquely. This proves both bijections. Since  $U$  has dimension two and the quotient has dimension  $\delta_2(L)$ , counting its nonzero classes proves the counting formula.

Finally suppose that  $m > 1$  is odd and  $\delta_2(L) > 0$ . The preceding construction gives an order- $m$  subsquare. Two distinct rows inside it induce a fixed-point-free permutation preserving its  $m$  columns. Such a permutation on an odd set has a nontrivial odd cycle, contradicting an F row view. The two parastrophes give the same conclusion for either other view, which proves the rank formula. The exception  $m = 1$  is real: the order-2 FFF square has four singleton subsquares and  $\delta_2 = 1$ .  $\square$

The complete finite audit independently constructs both sides of the bijection on 9778 tables and obtains exact set equality in every case. In particular, the 230 order-8 FFF representatives have

$$\delta_2 = 0, 1, 2, 3 \quad \text{with multiplicities} \quad 63, 156, 10, 1,$$

which explains their half-order-subsquare distribution  $0^{63}4^{156}12^{10}28^1$ . A hypothetical order-10 FFF square must have binary line rank 28 and no order-5 subsquare. This condition is not sufficient: all 135 tracked order-10 diagnostic tables already satisfy it, none is FFF, and that corpus is not exhaustive.

The same kernel equation has a uniform interpretation over every prime field, now in terms of quotient homotopies rather than individual subsquares. Moorhouse's character formula for loop-coordinatized 3-nets gives the general identity  $\text{rank}_p M_L = 3n - 2 - \dim \text{Hom}(G, \mathbb{F}_p)$  and identifies its defect with elementary-abelian quotient characters [Moo91, Theorem 4.2]. We record a direct three-sorted proof in our notation because the individual cyclic quotient is the input needed for the FFF corollary below. The author's publication page flags an erratum for the original article; the reference entry therefore points to the author-hosted article copy and the separate erratum listing. The direct proof given here, rather than an unexamined appeal to the original pagination, is the logical basis for the specialization used below.

**Theorem 4.4** (Prime-field line quotients). *Let  $p$  be prime and let  $(a, b, c) \in \ker_{\mathbb{F}_p} M_L^T$  be nonconstant modulo the two-dimensional constant subspace. Then  $p \mid n$ , each of  $a, b, c$  takes every value exactly  $n/p$  times, and their fibres satisfy*

$$L(R_x, C_y) \subseteq S_{-x-y} \quad (x, y \in \mathbb{F}_p).$$

*After a suitable isotopy these three fibre partitions become one quasigroup congruence with  $p$  blocks of size  $n/p$  and quotient isotopic to  $C_p$ .*

*Over characteristic zero,  $M_L$  has rank  $3n - 2$ . Every prime dividing a nonzero Smith invariant factor of the integer matrix  $M_L$  divides  $n$ .*

*Proof.* If one of  $a, b, c$  is constant, fixing a line in either other sort and using Latinness makes the remaining two constant. Let  $\mu_b(t)$  and  $\mu_c(t)$  be value multiplicities. Since every row permutes the symbols, equation (3), now read in  $\mathbb{F}_p$ , gives

$$\mu_c(t) = \mu_b(-a_r - t).$$

Comparing rows with  $a_r - a_{r'} \neq 0$  makes  $\mu_b$  invariant under a nonzero translation. That translation generates the additive group of  $\mathbb{F}_p$ , so  $\mu_b$  is uniform. Thus  $p \mid n$  and both  $b, c$  are balanced; the column argument balances  $a$ . The displayed fibre rule is the kernel equation itself.

Choose bijections from one partitioned  $n$ -set to the row, column, and symbol sets, matching equally labelled fibres. Transporting  $L$  by these three bijections gives an isotope whose block operation is  $x * y = -x - y$  on  $\mathbb{F}_p$ . It is Latin and isotopic to addition, so block equivalence is compatible with multiplication and both divisions and is a quasigroup congruence.

In characteristic zero a finite nonempty multiset cannot be invariant under a nonzero translation, because its iterated translates would have infinite support. Hence only the two constant dependencies remain. Finally, if a prime divides a nonzero Smith invariant factor, reduction modulo that prime lowers the rank; the first part then forces the prime to divide  $n$ .  $\square$

An eight-worker audit checks this theorem over  $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$  on 9778 tables. It verifies all 1160 nonuniversal quotient classes and a separate implementation reconstructs 48890 modular ranks, with no errors.

We next use congruences in the quasigroup algebra with multiplication and the two division operations. All congruence blocks have equal size and form a Latin quotient.

**Theorem 4.5** (Congruence obstructions). *Let  $L$  be a finite Latin square.*

- (i) *If  $L$  has a congruence with odd block size greater than one, then  $\text{pat}(L) = \text{TTT}$ .*
- (ii) *If the congruence blocks have size two and the quotient is  $Q$ , then  $\text{pat}(L) = \text{pat}(Q)$  componentwise.*

*Proof.* For (i), choose distinct rows  $x, x'$  in one congruence block  $A$  and any column block  $B$ . The two left translations map  $B$  bijectively onto the same product block. Thus  $L_{x'}^{-1}L_x$  preserves  $B$  and is fixed-point-free there. Since  $|B|$  is odd, its restriction has a nontrivial odd cycle. Applying the same argument to the two relevant parastrophes proves the column and symbol bits.

For (ii), label every two-element block by  $(i, 0), (i, 1)$ , where  $i \in Q$ . The unique form of a binary Latin square gives a function  $f: Q^2 \rightarrow \mathbb{F}_2$  such that

$$(i, a) * (j, b) = (i * j, a + b + f(i, j)).$$

Rows in the same quotient block induce only transpositions inside the column fibres. For rows in different quotient blocks, the lift above a quotient cycle of length  $\ell$  is either two cycles of length  $\ell$  or one cycle of length  $2\ell$ , according to its binary voltage. Thus an ambient odd cycle can occur only above an odd quotient cycle.

Conversely, suppose quotient rows  $i, i'$  induce an odd cycle  $C$  of length  $\ell > 1$ . Above a quotient column  $j \in C$ , the lifted row permutation comparing  $(i, a)$  with  $(i', a')$  has the form

$$(j, b) \mapsto (\pi(j), b + a + a' + f(i, j) + f(i', \pi(j))).$$

Its voltage around  $C$  is  $a + a' + S$ , because  $\ell$  is odd, where  $S$  is independent of  $a, a'$ . Choose  $a + a' = S$ . The voltage is then zero, so the lift of  $C$  consists of two cycles of the same odd length  $\ell$ . Hence a quotient row witness lifts to an ambient row witness. This proves equality of the row bits. The same complete argument in the parastrophes proves equality of the column and symbol bits.  $\square$

**Corollary 4.6** (Order- $2p$  simplicity). *Let  $p$  be an odd prime. Every quasigroup of order  $2p$  with at least one  $F$  view is congruence-simple, and every isotope of it is congruence-simple.*

*Proof.* A nontrivial congruence has block-size/quotient-order pair  $(p, 2)$  or  $(2, p)$ . The first is excluded by Theorem 4.5(i). In the second, the quotient has pattern TTT by Proposition 3.1, so (ii) excludes it. Both congruence cases force TTT, contradicting even one  $F$  view. Finally, the full view pattern is isotopy invariant by Proposition 2.9, and the same argument applies to every isotope.  $\square$

**Corollary 4.7** (Order- $2p$  line-incidence saturation). *Let  $p > 1$  be an odd prime and let  $L$  have order  $2p$ . If even one view of  $L$  is  $F$ , then  $M_L$  has rank  $6p - 2$  over every field, all its nonzero Smith invariant factors are 1, and*

$$\mathbb{Z}^{6p} / \text{im}_{\mathbb{Z}} M_L \cong \mathbb{Z}^2.$$

*In particular, this holds for every hypothetical order-10 FFF square.*

*Proof.* By Theorem 4.4, a torsion prime must divide  $2p$ . A characteristic-two defect yields, after isotopy, a congruence with two blocks of odd size  $p$ , so Theorem 4.5(i) gives pattern TTT. A characteristic- $p$  defect yields blocks of size two; part (ii) identifies the pattern with that of the odd-order quotient, again TTT by Proposition 3.1. Isotopy invariance contradicts the assumed F view in either case. Thus no Smith prime remains, proving saturation and the displayed cokernel.  $\square$

Thus a hypothetical order-10 FFF square must be isotopically congruence-simple. This removes every congruence-imprimitive quasigroup construction, but it does not rule out simple quasigroups and therefore does not decide order 10.

The binary equality iterates through congruence series.

**Corollary 4.8** (Prime-factor congruence series). *Suppose that a quasigroup admits a chain of congruence quotients ending in the one-element quasigroup, with every congruence block size prime. It is FFF if all those primes are 2, and it has pattern TTT if at least one is odd.*

*Proof.* A binary step preserves the entire pattern by Theorem 4.5(ii). At an odd step, Theorem 4.5(i) gives TTT independently of the lower quotient. Iterating proves both statements.  $\square$

This proves the power-of-two conjecture for this congruence-decomposable class, but leaves congruence-simple quasigroups outside its scope.

The same theorem also gives a census-free nongroup example.

**Theorem 4.9** (Explicit nongroup FFF loop). *On  $\mathbb{Z}_4 \times \mathbb{F}_2$ , define*

$$(i, a) * (j, b) = (i + j, a + b + \mathbf{1}_{(i,j)=(1,1)}),$$

*with coordinates taken modulo 4 and 2, respectively. This is an FFF loop of order 8 that is not group-isotopic.*

*Proof.* The identity is  $(0, 0)$ , and the first-coordinate fibres form a congruence with quotient  $C_4$ . The quotient is FFF by Corollary 3.3, so binary pattern equality makes the loop FFF. It is not associative because

$$((1, 0) * (1, 0)) * (2, 0) = (0, 1), \quad (1, 0) * ((1, 0) * (2, 0)) = (0, 0).$$

At the identity base row, the basis family in Proposition 3.4 is the family of left translations. Closure would force  $L_x L_y = L_{x*y}$  and hence associativity. The displayed failure is therefore a closure failure, so the loop is not group-isotopic.  $\square$

## 5 Direct products

Let  $L$  be a Latin square on  $X$  and  $K$  a Latin square on  $Y$ . Their direct product  $P = L \times K$  is the Latin square on  $X \times Y$  defined by

$$P((x, y), (x', y')) = (L(x, x'), K(y, y')).$$

**Lemma 5.1** (Product-cycle formula). *Let  $\pi$  and  $\tau$  be permutations. If  $x$  lies on a  $\pi$ -cycle of length  $m$  and  $y$  lies on a  $\tau$ -cycle of length  $n$ , then  $(x, y)$  lies on a cycle of length  $\text{lcm}(m, n)$  under  $\pi \times \tau$ .*

*Proof.* The pair returns after  $t > 0$  steps exactly when both  $m \mid t$  and  $n \mid t$ . The least such  $t$  is  $\text{lcm}(m, n)$ .  $\square$

**Theorem 5.2** (Pattern product theorem). *For arbitrary Latin squares  $L$  and  $K$ ,*

$$\text{pat}(L \times K) = \text{pat}(L) \text{ OR } \text{pat}(K)$$

*componentwise.*

*Proof.* Rows of  $P$  are indexed by  $(a, \alpha)$ , and their row maps are  $r_{(a, \alpha)}^P = r_a^L \times r_\alpha^K$ . Hence the induced permutation between  $(a, \alpha)$  and  $(b, \beta)$  is

$$\rho_{(a, \alpha), (b, \beta)}^P = \rho_{a, b}^L \times \rho_{\alpha, \beta}^K.$$

If one coordinate of the two product rows is equal, the corresponding factor is the identity. The identical calculation for column maps gives  $\kappa^P = \kappa^L \times \kappa^K$ .

For the symbol view,

$$s_{(u, \eta)}^P(c, d) = (s_u^L(c), s_\eta^K(d)),$$

and therefore  $\sigma^P = \sigma^L \times \sigma^K$ . Thus the same argument applies in all three views.

If a factor has a nontrivial odd cycle in one view, keep the other factor line fixed. The identity contributes cycle length 1, so the product contains the same odd cycle. Conversely, suppose a product induced permutation has an odd cycle of length  $\ell > 1$ . By Lemma 5.1,  $\ell = \text{lcm}(m, n)$  for factor cycle lengths  $m, n$ . Both are odd, and at least one is greater than one. That factor already has a nontrivial odd cycle in the same view. This proves the Boolean OR formula in each view.  $\square$

**Corollary 5.3.**  *$L \times K$  is FFF if and only if both  $L$  and  $K$  are FFF. In particular, product with a finite 2-group table preserves the full pattern of the other factor.*

*Proof.* The first assertion is the all- $F$  case of Theorem 5.2. For the second, combine the same theorem with Corollary 3.3.  $\square$

**Theorem 5.4** (Group-isotopy product theorem).  *$L \times K$  is group-isotopic if and only if both  $L$  and  $K$  are group-isotopic.*

*Proof.* At a product base row  $(b, c)$ , the basis family factors as

$$H_{L \times K}(b, c) = H_L(b) \times H_K(c).$$

If both factor families are closed, their Cartesian product is closed. Conversely, both factor families contain the identity. If the product family is closed, then  $(h_1, \text{id})(h_2, \text{id})$  lies in it for all  $h_1, h_2 \in H_L(b)$ , proving closure of  $H_L(b)$ ; using  $(\text{id}, k_i)$  proves closure of  $H_K(c)$ . Apply Proposition 3.4.  $\square$

**Corollary 5.5.** *Let  $L$  be a non-group-isotopic FFF Latin square of order  $n$ . For every  $m \geq 0$ , the product  $L \times C_2^m$  is a non-group-isotopic FFF Latin square of order  $n2^m$ .*

*Proof.* The table of  $C_2^m$  is FFF by Corollary 3.3; apply Corollary 5.3 and Theorem 5.4.  $\square$

## 6 Computational results

The following statements are exhaustive finite computations. Their frozen inputs, programs, commands, outputs, and hashes are specified in Section 7.

**Computational Theorem 6.1** (Small orders). *The unique reduced Latin square of order 2 is FFF, and all four reduced Latin squares of order 4 are FFF. None of the 9408 reduced Latin squares of order 6 is FFF. More precisely, their patterns are*

$$6600\text{ TTT}, \quad 936\text{ TFF}, \quad 936\text{ FTF}, \quad 936\text{ FFT}.$$

The exact-cover generator loads no cached square list. Every generated square was checked to be Latin and reduced, all 9408 table strings were distinct, and the generated set agrees exactly with the ANU reduced-order-6 data file [McK]. Proposition 2.9 therefore extends nonexistence from reduced squares to all Latin squares of order 6.

**Computational Theorem 6.2** (Order-8 pattern census). *Across the 283657 order-8 main-class representatives, the pattern distribution is*

| <i>Pattern</i> | <i>Count</i> | <i>Pattern</i> | <i>Count</i> | <i>Pattern</i> | <i>Count</i> | <i>Pattern</i> | <i>Count</i> |
|----------------|--------------|----------------|--------------|----------------|--------------|----------------|--------------|
| FFF            | 230          | FFT            | 81           | FTF            | 40           | FTT            | 315          |
| TFF            | 133          | TFT            | 209          | TTF            | 232          | TTT            | 282417       |

*Among the 230 FFF main classes, exactly 5 are group-isotopic and 225 are not group-isotopic.*

**Computational Theorem 6.3** (Subsquare census of the FFF classes). *Among the 230 FFF representatives, the number of intercalates ranges from 2 to 112. Each endpoint is attained by exactly one representative: local source line 10776 (ANU record 10775) is the unique minimum class and is not group-isotopic, while local line 1 (ANU record 0) is the unique maximum class and is group-isotopic. Local physical lines are numbered from one; ANU records are numbered from zero. No representative contains an order-3 subsquare. The distribution of the number of order-4 subsquares is*

$$0^{63}, \quad 4^{156}, \quad 12^{10}, \quad 28^1,$$

*where exponents count representatives.*

The subsquare validator finds no violation of Theorem 4.1. The multiples of four in the last distribution also follow directly: an order- $m$  subsquare inside an order- $2m$  square forces the three complementary blocks to be Latin subsquares, so half-order subsquares occur in complementary quartets.

The ordered non-symmetric labels refer to the representatives in the cited ANU file. Parastrophy permutes the three bits, so labels such as FFT are not themselves main-class invariants. The FFF count, the existence of every pattern, and the group-isotopy split are unaffected.

The scan tests every unordered pair of lines in all three views. A separate closure-family computation reproduces the 5/225 split using Proposition 3.4. The frozen ANU input is an exact byte match to the file served by the cited URL on August 9, 2026.

**Corollary 6.4.** *For every  $m \geq 3$ , Latin squares of order  $2^m$  realize all eight patterns. Every such order also contains a non-group-isotopic FFF Latin square.*

*Proof.* Computational Theorem 6.2 supplies one order-8 square of each pattern. Take direct products with the FFF group table of  $C_2^{m-3}$  and use Theorem 5.2. For the second assertion, start from the explicit loop in Theorem 4.9; non-group-isotopy is preserved by Theorem 5.4.  $\square$

This is an existence result only. It neither counts higher-order main classes nor asserts that products of distinct order-8 classes remain distinct. **Historical reproducibility record.** This section describes the underlying research collection, not a promise that every named artifact is in this public export. See the dossier scope statement and the public reproducibility guide for availability.

## 7 Computational reproducibility

The historical research collection separates proofs, frozen data, programs, commands, logs, and SHA-256 manifests. A companion reproducibility quickstart supplies the complete command-to-output map. The compact paper uses the following packages.

**Trade parity** `repro_runs/2026-08-10_paper_symbol_sign_audit/`. The exact control checks 1028307 row-, column-, and symbol-view trades on all reduced Latin squares of orders 2, 4, and 6, with zero failures. The proof, not this computation, establishes the general theorem.

**Small orders** `repro_runs/2026-04-26_small_order_witness_reruns/`. The exact-cover generator freshly enumerates 1, 4, and 9408 reduced squares of orders 2, 4, and 6. Every table is checked Latin, reduced, and unique. The generated order-6 set agrees exactly with the official ANU file.

**Order-8 census** `repro_runs/2026-04-26_order8_core/`. The frozen 5140393-byte input has SHA-256 `e5980ff5bc4ac46cbbefd1f1333258d0905724c42a976c4afe61ea501a4ccf87`. The theorem-relevant output `results/order8_recheck_fullscan_results.json` has SHA-256 `10f2f43c8dcaa8ec93f83abf6b768efa47f909a9c8623d4bc648e6da097ccd39`.

**Group-isotopy closure** `repro_runs/2026-04-26_order8_closure_reconstruction/`. This independently reconstructs the basis-family closure scan and reproduces the 5/225 split.

**Direct products** `repro_runs/2026-04-26_direct_product_fff/`. The validator checks the product-permutation formula in all three views and tests the group-isotopy equivalence on group and non-group examples.

**Subsquare and congruence obstructions** `repro_runs/2026-08-09_fff_subsquare_congruence_obstructions/`. The package scans all subsquares of the 230 frozen FFF representatives, exhausts the binary twists used as controls, and validates the explicit twisted order-8 loop.

**Binary line kernel and half-order subsquares** `repro_runs/2026-08-11_fff_binary_homotopy_subsquares/`. The package compares the subsquares constructed from every nonuniversal binary line dependency with an independent direct enumeration on 9778 tables. All predicted and observed sets agree exactly, and the semantic verifier reproduces the frozen order-8 distribution with zero errors.

**Prime-field line quotients** `repro_runs/2026-08-11_fff_prime_field_line_quotients/`. The audit checks five prime fields on 9778 tables, verifies all 1160 nonuniversal quotient classes, and independently reconstructs 48890 modular ranks with zero errors. The theorem itself is proved separately.

**External-input release audit** `repro_runs/2026-08-09_paper_release_audit/`. It downloads the cited ANU and Wanless inputs afresh and verifies exact byte matches to the frozen files. As with any computational theorem based on an external census, completeness of the published enumeration remains a cited scientific dependency.

Every package contains a README, exact commands, environment record, input and output manifests, and a package-level SHA-256 manifest. The command

```
python3 repro_runs/2026-08-10_paper_symbol_sign_audit/scripts/audit_manuscript_computational_claims.py
```

cross-checks the computational statements in the long companion manuscript; the three computational theorems retained here are a subset of that audit.

Solver binaries, virtual environments and non-research source material are not included in the public export. No timeout, partial solver model, or order-10 diagnostic is used as evidence in this compact paper.

## 8 Discussion and open problems

Kobayashi and Nakamura studied the stricter condition that every pair of factors in a 1-factorization of  $K_{n,n}$  forms only 4-cycles; such a factorization exists precisely in power-of-two orders [KN94]. Our F condition permits arbitrary even induced cycle lengths, but requires the condition simultaneously in all three coordinate views. The neighbouring classification, the odd-order obstruction, the order-6 computation, and the power-of-two constructions motivate the following conjecture.

**Conjecture 8.1** (Power-of-two FFF conjecture). For  $n > 1$ , an FFF Latin square of order  $n$  exists if and only if  $n$  is a power of two.

The positive direction follows from Corollary 3.3. The negative direction is proved for odd  $n$  by Proposition 3.1 and computed for  $n = 6$  by Computational Theorem 6.1. The accompanying complete report excludes order 10 by exhaustive cycle-palette master searches. The first undecided even non-power-of-two order is therefore 12.

Corollary 4.6 provides a structural reduction at order 10: every order-10 FFF candidate would have to be congruence-simple in every isotope. Hence no central extension, binary extension, or other congruence-imprimitive construction can settle the positive direction. This compact-paper theorem alone does not exclude simple quasigroups; the companion report’s exhaustive computation does.

Three concrete questions remain:

1. Decide the conjecture at order 12 or prove a uniform obstruction for even non-powers of two.
2. Find an order-independent obstruction coupling all three link factorizations more strongly than their common intercalate count.
3. Determine when direct products preserve distinctions between isotopy or main classes. No such injectivity is asserted here.

These questions concern one local odd-cycle obstruction. They must not be conflated with the global Alon–Tarsi conjecture, whose resolution would require substantially more than existence or absence of this trade primitive [AT92].

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